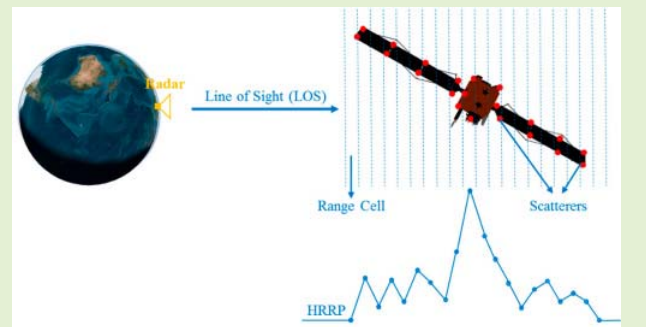


Online Factor Analysis Model With Kullback–Leibler Constraint for Satellite Target Recognition

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Abstract—Satellite target data are usually obtained over a long period in streaming mode rather than at one time. In satellite target recognition, most of existing methods are in offline mode. With the arrival of new data, the offline learning methods have to retrain a new model with both historical and new data. However, the repetitive retraining not only leads to a huge burden for storage but also results in poor real-time performance, due to the storage and repeated calculation of all historical data. To address this problem, this paper presents a novel satellite target recognition method based on online factor analysis model with Kullback–Leibler constraint (OnFA-KL model). On the one hand, the parameters of OnFA-KL model can be updated by only using new data without revisiting the past data; on the other hand, a Kullback–Leibler (KL) divergence term is introduced to preserve the previous learned knowledge of the past data, thus alleviating the catastrophic forgetting. In addition, owing to the streaming mode of the satellite target data, streaming variational Bayesian (SVB) algorithm is employed to infer the parameters of our OnFA-KL model. In the simulation experiments, the data of five satellite targets we use are generated by using real orbit parameters and computer aided design (CAD) models. Numerical experiments show that compared with the offline learning method, the proposed method achieves comparable recognition performance and improves the learning efficiency.

Index Terms—Factor analysis (FA), online learning, radar automatic target recognition (RATR), satellite target, high-resolution range profile (HRRP).



I. INTRODUCTION

SATELLITE target recognition, as a primary work of space target recognition [1], [19]–[21], [37], has increasingly attracted attention in the field of space target surveillance system. Due to the rapid development of wideband radar, lots of useful data can be available, such as high-resolution range profile (HRRP) [1]–[5], [12], [13], [15], [16], [22] and Inverse synthetic aperture radar (ISAR) image [6]–[10], even if the satellite targets are too small or too far. In contrast with the ISAR images, based on the property of easy acquisition and processing of the HRRPs, HRRP-based radar automatic target recognition (RATR) has been extensively studied. An HRRP is composed of the amplitude of coherent summations of

complex returns from target scatterers in each range cell, and it provides rich target information along the radar line-of-sight, such as scatterer distribution and structural characteristics.

In HRRP-based satellite target recognition, most existing methods are in offline mode. In [19], Liu *et al.* utilized HRRP even rank central feature for nearest neighbor fuzzy classifier in satellite target recognition. Xiankang *et al.* [20] took advantage of Back-Propagation Neural Network for HRRP-based satellite target recognition, in which Back-Propagation algorithm was proposed to process even rank central moments of HRRP. A deep learning model [21] which was composed of a multi-layer Gated Recurrent Unit Neural Network as a deep feature extractor and a linear Support Vector Machine as a classifier was applied in satellite target recognition.

In real scenarios, satellite target data with multi-circle are usually obtained continuously in streaming mode in order of circles over a long period, often several years or even decades. During the period of data acquisition, the abovementioned offline models have to discard the previous learned knowledge and retrain the model from scratch when new data come in, i.e., the offline learning methods require repetitive training with all obtained data, including all storing historical data and

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new data, thus weakening real-time performance. In the field of space target surveillance system, the main task is to systematically and continuously observe and collect information about the space target for discovery, recognition, forecasting and etc. Hence, we should process and analyze the relevant information of the space target in time for providing real-time and effective support of the space activity, orbit security, attack interception, and etc. Based on the above, an ideal satellite target recognition system is expected to update the model in an online learning mode in space target surveillance system. In contrast to the offline learning methods, the online learning methods update the current training model in real time like the manner of human cognition, rather than retraining a new model. Some previous works have shown that some online learning methods successfully perform in many applications. A fast algorithm based on online Principal Component Analysis (OnPCA) model was proposed in [17], which was inspired by an incremental algorithm for finding the singular value decomposition. Dang *et al.* developed an online nonnegative matrix factorization (OnNMF) model [38], which is a feature-updating method. With the arrival of the training data, the base matrix and the encoding matrix can be updated in online learning mode, and the classification is achieved by support vector machine (SVM). In [18], Li *et al.* extended beta-process factor analysis model into online mode for processing large-scale datasets. The parameters were updated via stochastic variational inference (SVI) algorithm [28]–[31], [35], which is a stochastic optimization algorithm for mean-field variational inference [26]. The main idea of SVI algorithm is to combine the stochastic gradient descent with variational inference, and it has succeeded in processing large-scale datasets of online scenarios. Note that SVI requires to specified the size of the dataset as a priori, which is an inappropriate assumption for streaming data. The above beta-process factor analysis model in online mode is referred to as online stochastic factor analysis (OnSFA) model in the following. However, as more training data are gathered, these online learning methods suffer from a serious catastrophic forgetting problem, namely, adapting a model to new data results in severe performance degradation on past data. Consequently, the recognition performance is badly affected.

Since statistical models, such as principal component analysis (PCA) model [16], Gaussian model [13], probabilistic PCA model [16], the two distribution compounded model comprising Gamma distribution and Gaussian mixture distribution [4], and factor analysis (FA) model [3], [16], are effective for HRRP-based target recognition, we aim to develop a kind of online statistical model. Among these statistical models, Du *et al.* indicated that the FA model [16] performs well in describing the statistical characteristics of the HRRP data, so as to achieve favorable performance in HRRP-based statistical recognition. In this paper, we develop a novel satellite target recognition method based on online factor analysis model with Kullback-Leibler constraint (OnFA-KL model). The satellite target recognition method based on OnFA-KL model is a statistical recognition method, target-aspect sensitivity [3], [11], [16] must be considered. Some previous works [3], [16] have shown that this problem can be addressed

by the aspect-frames partition method, and suppose HRRPs within the same aspect-frame obey the same distribution, whereas the HRRPs from different aspect-frames obey the different distributions.

Considering that some satellite targets are not geosynchronous satellites, the relative positions between radar and the satellite target are different in each circle; thus, the ranges of target-aspect angles of different circles may be different. With the arrival of new data, the new data may belong to existing aspect-frames or new aspect-frames. For the existing aspect-frames, the parameters of our OnFA-KL model can be updated in online mode without revisiting the past data, which improves the learning efficiency and saves the cost of storage. Moreover, a Kullback-Leibler (KL) divergence term is introduced to preserve the previous learned knowledge of the past data. Our OnFA-KL model adds the KL constraint, which restrains the parameters updated from two adjacent incoming data to be close, to prevent the updated model to deviate too much from the previous model. The previous learned knowledge of the past data can be preserved, so that our method can alleviate catastrophic forgetting problem. For the new aspect-frames, because of the assumption that the data from different aspect-frames have different distributions, new submodels are constructed with the incoming data of new aspect-frames to describe the distribution of new aspect-frames precisely, instead of directly updating the existing model. In addition, efficient inference is performed via streaming variational Bayesian (SVB) algorithm [32] for OnFA-KL model. Unlike the traditional variational Bayesian (VB) algorithm used for offline learning methods, SVB is an online learning algorithm which handles an infinite amount of streaming data by exploiting the sequential nature of Bayes theorem to recursively update the parameters in online learning mode. The main idea of SVB is to treat the posteriors trained at the previous moment as the priors of the current moment. In general, SVB algorithm is used in streaming mode for processing the single pass of the data. Compared with SVI algorithm, SVB algorithm outperforms SVI algorithm in streaming mode, where the past data could not be repetitively utilized. The reason is that the SVI learning rate requires to be specifically tuned by revisiting the data multiple times to obtain good performance.

In our experiments, the satellite HRRP data are obtained by simulation, which uses real orbit parameters and computer aided design (CAD) models of five satellite targets. Experiments on this data show that compared with the offline learning method, the proposed method achieves comparable recognition performance and improves the learning efficiency.

In summary, the main contributions of this work are the following:

- We propose a novel OnFA-KL model for online HRRP-based satellite target recognition. The parameters of our OnFA-KL model can be updated by only using new data without revisiting the past data. Compared with the offline learning method, our method improves the learning efficiency and reduces the cost of storage.
- A KL divergence term is introduced into our OnFA-KL model to preserve the previous learned

knowledge of the past data for alleviating catastrophic forgetting.

- The parameters of our OnFA-KL model are updated via SVB algorithm. Compared with existing online FA model (i.e., OnSFA model) being updated via SVI, SVB is more suitable for streaming data and more accurate.

The remainder of the paper is arranged as follows. Section II is a brief overview of the related work. Then, our method is introduced in Section III. Further, the detailed experiments are presented to evaluate the effectiveness and efficiency of our method in Section IV, followed by conclusions in Section V.

II. RELATED WORK

In this section, we give an introduction about the related work. Firstly, an overview of traditional offline factor analysis model is given. Then, the relationship between traditional learning methods and our proposed OnFA-KL model is introduced, which helps readers understand the proposed method more easily.

Traditional FA model [14], [34] is a linear Gaussian generative model for modeling the correlations amongst high-dimensional data in terms of a low-dimensional latent variable. Denote the HRRP samples in the m th aspect-frame of the target c by $\mathbf{X}^{c,m} = \{\mathbf{x}_i^{c,m}\}_{i=1}^N$ ($\mathbf{x}_i^{c,m} \in \mathbb{R}^P$), where $c = 1, 2, \dots, C$ indexes the class label, $m \in \{1, \dots, M_c\}$ denotes the frame index with M_c being the number of aspect-frames of c th target, and N represents the size of this frame. In FA model, it is assumed that a latent variable is linearly mapped and added an observation noise to generate $\mathbf{x}_i^{c,m}$, and $\mathbf{x}_i^{c,m}$ can be represented as follows:

$$\begin{aligned} \mathbf{x}_i^{c,m} &= \mathbf{A}^{c,m} \mathbf{s}_i^{c,m} + \boldsymbol{\mu}^{c,m} + \boldsymbol{\varepsilon}_i^{c,m} \\ \mathbf{s}_i^{c,m} &\sim N(\mathbf{0}, \mathbf{I}_K) \\ \mathbf{a}_j^{c,m} &\sim N(\mathbf{0}, \mathbf{I}_K) \\ \boldsymbol{\varepsilon}_i^{c,m} &\sim N(\mathbf{0}, (\boldsymbol{\gamma}^{c,m})^{-1}), \quad \gamma_j^{c,m} \sim Ga(a_0, b_0), \end{aligned} \quad (1)$$

where $\boldsymbol{\mu}^{c,m}$ is the mean vector of the data in the m th aspect-frame of the target c ; $\mathbf{s}_i^{c,m}$ is the latent variable that obeys the Gaussian distribution as $\mathbf{s}_i^{c,m} \sim N(\mathbf{0}, \mathbf{I}_K)$ with $\mathbf{I}_K$ denoting $K \times K$ dimensional identity matrix and K being the number of factors ($P \gg K$); $\mathbf{a}_j^{c,m}$ is a column vector corresponding to the j th row of the factor loading matrix $\mathbf{A}^{c,m}$; the noise variable $\boldsymbol{\varepsilon}_i^{c,m}$ is assumed as $\boldsymbol{\varepsilon}_i^{c,m} \sim N(\mathbf{0}, (\boldsymbol{\gamma}^{c,m})^{-1})$, $\boldsymbol{\gamma}^{c,m}$ is a diagonal matrix with $\{\gamma_j^{c,m}\}_{j=1}^P$ being its diagonal elements and $\gamma_j^{c,m} \sim Ga(a_0, b_0)$; a_0, b_0 are preset hyper-parameters, and $Ga(\cdot)$ denotes the Gamma distribution. Then, $\mathbf{x}_i^{c,m}$ is also Gaussian distributed, and we have $\mathbf{x}_i^{c,m} \sim N(\boldsymbol{\mu}^{c,m}, \mathbf{A}^{c,m}(\mathbf{A}^{c,m})^T + (\boldsymbol{\gamma}^{c,m})^{-1})$. In this paper, we prefer this FA model as offline FA (OffFA) model. The graphical structure of the OffFA model is in Fig. 1 for detailed illustration.

In general, the VB inference [23]–[25], [34], is commonly adopted to infer the posterior distributions of all latent variables and parameters $\boldsymbol{\tau}^{c,m} = \{\mathbf{S}^{c,m}, \mathbf{A}^{c,m}, \boldsymbol{\gamma}^{c,m}\}$ in the OffFA model, given training data $\mathbf{X}^{c,m}$ and the hyper-parameters $\boldsymbol{\Psi}^{c,m} = \{a_0, b_0\}$. The aim of variational inference can be

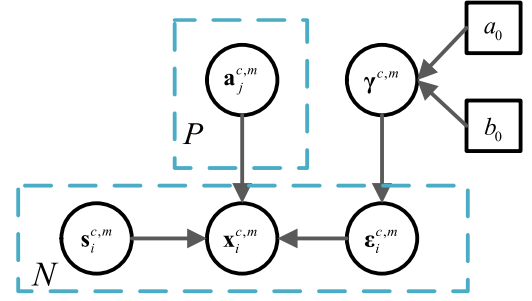


Fig. 1. Graphical structure of OffFA model. The corresponding generative process is depicted in (1).

regarded as maximizing the lower bound, which can be expressed as:

$$\begin{aligned} LB &= \int \ln \frac{p(\mathbf{X}^{c,m}, \boldsymbol{\tau}^{c,m} | \boldsymbol{\Psi}^{c,m})}{q(\boldsymbol{\tau}^{c,m})} q(\boldsymbol{\tau}^{c,m}) d\boldsymbol{\tau}^{c,m} \\ &= \ln p(\mathbf{X}^{c,m} | \boldsymbol{\Psi}^{c,m}) + KL(p(\boldsymbol{\tau}^{c,m} | \mathbf{X}^{c,m}, \boldsymbol{\Psi}^{c,m}) \| q(\boldsymbol{\tau}^{c,m})), \end{aligned} \quad (2)$$

where $q(\boldsymbol{\tau}^{c,m})$ is the unknown variational distribution of $\boldsymbol{\tau}^{c,m}$. $KL(p(\cdot) \| q(\cdot))$ is the KL divergence between $p(\cdot)$ and $q(\cdot)$, which is constantly nonnegative. The variational method iteratively updates each parameter alternatively until the lower bound is maximized.

Traditional OffFA model has achieved excellent performance, yet often run in a batch (offline) learning mode, where a collection of training data is given in advance. In real-world scenarios, the data are obtained in streaming mode. Thus, it is required to store massive streaming data (including historical data and new data) in memory, which is infeasible in many cases. Moreover, we have to retrain traditional OffFA model from scratch with all obtained data when new data come in, and this repetitive retraining operation lacks of real-time performance. Therefore, traditional OffFA model has two main drawbacks: 1) It may lead to low efficiency in data processing, i.e., poor real-time performance; 2) With the increase of data, more and more storage of computer is required, and the data may be too large to process on account of the limitation of computer capability. Based on these drawbacks, we develop OnFA-KL model. OnFA-KL model utilize new data to update the model previously learned from historical data without revisiting the historical data. Compared with traditional OffFA model, our OnFA-KL model needs less memory and can respond data faster, thus enhancing real-time performance. In addition, our OnFA-KL model adds a KL divergence term to alleviate catastrophic forgetting problem.

III. SATELLITE TARGET RECOGNITION METHOD BASED ON ONLINE FACTOR ANALYSIS MODEL WITH KULLBACK-LEIBLER CONSTRAINT

In this section, we develop a satellite target recognition method based on OnFA-KL model. Since our method is in online mode with streaming updates, the above VB algorithm is no longer applicable, and SVB algorithm [32] is more suitable for this streaming case.

A. Online Factor Analysis Model With Kullback-Leibler Constraint

We define HRRP samples in the m th aspect-frame of the target c obtained at time t as $\mathbf{X}^{c,m,t} = \{\mathbf{x}_i^{c,m,t}\}_{i=1}^{N_t}$ with N_t being the size of mini-batch $\mathbf{X}^{c,m,t}$, and superscript t denotes the variables corresponding to time t . The data in streaming mode are represented by $\mathbf{X}^{c,m} = \{\mathbf{X}^{c,m,0}, \mathbf{X}^{c,m,1}, \mathbf{X}^{c,m,2}, \dots\}$. Note that the amount of data in a stream might be infinite. In accordance with the key idea of SVB, which treats the posteriors of global variables trained with the past $t-1$ mini-batches as the priors of the t th mini-batch, the OnFA-KL model and all parameters' prior in the model at time t can be expressed as

$$\begin{aligned} \mathbf{x}_i^{c,m,t} &= \mathbf{A}^{c,m,t} \mathbf{s}_i^{c,m,t} + \boldsymbol{\mu}^{c,m,0:t} + \boldsymbol{\varepsilon}_i^{c,m,t} \\ \mathbf{s}_i^{c,m,t} &\sim N(\mathbf{0}, \mathbf{I}_K) \\ \mathbf{a}_j^{c,m,t} &\sim N\left(\mathbf{m}_{\mathbf{a}_j^{c,m,t-1}}, \boldsymbol{\Sigma}_{\mathbf{a}_j^{c,m,t-1}}\right) \\ \boldsymbol{\varepsilon}_i^{c,m,t} &\sim N\left(\mathbf{0}, (\boldsymbol{\gamma}^{c,m,t})^{-1}\right), \boldsymbol{\gamma}_j^{c,m,t} \\ &\sim Ga\left(a_{\boldsymbol{\gamma}_j^{c,m,t-1}}, b_{\boldsymbol{\gamma}_j^{c,m,t-1}}\right), \end{aligned} \quad (3)$$

where $\boldsymbol{\mu}^{c,m,0:t}$ is the mean vector of the all data obtained until time t in the m th aspect-frame of the target c ; the calculation of $\boldsymbol{\mu}^{c,m,0:t}$ can be implemented as $\boldsymbol{\mu}^{c,m,0:t} = \left(\boldsymbol{\mu}^{c,m,0:t-1} \sum_{r=0}^{t-1} N_r + \sum_{i=1}^{N_t} \mathbf{x}_i^{c,m,t}\right) / \sum_{r=0}^t N_r$; $\mathbf{s}_i^{c,m,t}$ is the latent variable; $\mathbf{a}_j^{c,m,t}$ is a column vector corresponding to the j th row of the factor loading matrix $\mathbf{A}^{c,m,t}$ with $\mathbf{a}_j^{c,m,t} \sim N\left(\mathbf{m}_{\mathbf{a}_j^{c,m,t-1}}, \boldsymbol{\Sigma}_{\mathbf{a}_j^{c,m,t-1}}\right)$, and $\mathbf{m}_{\mathbf{a}_j^{c,m,t-1}}, \boldsymbol{\Sigma}_{\mathbf{a}_j^{c,m,t-1}}$ are the mean vector and the covariance matrix of the posterior distribution of $\mathbf{a}_j^{c,m,t-1}$ respectively, which are learned from the past $t-1$ mini-batches; the noise variable $\boldsymbol{\varepsilon}_i^{c,m,t}$ is assumed as $\boldsymbol{\varepsilon}_i^{c,m,t} \sim N\left(\mathbf{0}, (\boldsymbol{\gamma}^{c,m,t})^{-1}\right)$ with $(\boldsymbol{\gamma}^{c,m,t})^{-1}$ being the covariance matrix of the prior distribution; $\boldsymbol{\gamma}^{c,m,t}$ is a diagonal matrix with $\{\gamma_j^{c,m,t}\}_{j=1}^P$ denoting its diagonal elements, and $\gamma_j^{c,m,t} \sim Ga\left(a_{\gamma_j^{c,m,t-1}}, b_{\gamma_j^{c,m,t-1}}\right)$; analogously, $a_{\gamma_j^{c,m,t-1}}, b_{\gamma_j^{c,m,t-1}}$ the hyperparameters of the posterior distribution for $\gamma_j^{c,m,t-1}$, which are learned from the past $t-1$ mini-batches. The factor loading matrix $\mathbf{A}^{c,m,t}$ and the parameter $\boldsymbol{\gamma}^{c,m,t}$ are global variables $\boldsymbol{\theta}$ which are not specific to the most recent mini-batch, and the latent variable $\mathbf{s}_i^{c,m,t}$ is local variable $\boldsymbol{\xi}^t$. Then, similar to Section II, we have $\mathbf{x}_i^{c,m,t} \sim N\left(\boldsymbol{\mu}^{c,m,0:t}, \mathbf{A}^{c,m,t} (\mathbf{A}^{c,m,t})^T + (\boldsymbol{\gamma}^{c,m,t})^{-1}\right)$.

Considering that the model is updated gradually without revisiting the past data, with the arrival of new data, the method in online mode suffers from catastrophic forgetting. Thus, a KL divergence term is introduced to preserve the previous learned knowledge of the past data. The KL divergence [27] is used to measure the closeness between two probability density functions (PDFs). The more similar two distributions are, the smaller KL divergence between

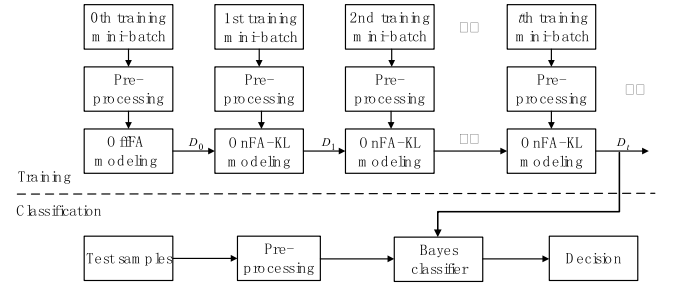


Fig. 2. The flow chart of our target recognition method based on OnFA-KL model, where the classification stage can be performed at any time.

them is. By adding the KL constraint term, the objective function of OnFA-KL model can be expressed as:

$$\begin{aligned} L(\boldsymbol{\theta}, \boldsymbol{\xi}^t) &= \int q(\mathbf{A}^{c,m,t}, \mathbf{S}^{c,m,t}, \boldsymbol{\gamma}^{c,m,t}) \\ &\ln \frac{p(\mathbf{X}^{c,m,t}, \mathbf{A}^{c,m,t}, \mathbf{S}^{c,m,t}, \boldsymbol{\gamma}^{c,m,t})}{q(\mathbf{A}^{c,m,t}, \mathbf{S}^{c,m,t}, \boldsymbol{\gamma}^{c,m,t})} d\mathbf{A}^{c,m,t} d\mathbf{S}^{c,m,t} d\boldsymbol{\gamma}^{c,m,t} \\ &- u \cdot KL\left(q(\mathbf{A}^{c,m,t}, \boldsymbol{\gamma}^{c,m,t}) \parallel q(\mathbf{A}^{c,m,t-1}, \boldsymbol{\gamma}^{c,m,t-1})\right), \end{aligned} \quad (4)$$

where the first term denotes the lower bound of logarithm marginal likelihood representing the degree of learning knowledge from the t th mini-batch; the second term is the KL constraint, which restrains global variables' posterior distributions learned from $\mathbf{X}^{c,m,t}$ and $\mathbf{X}^{c,m,t-1}$ to be similar; the weight value u is a tradeoff between the above two terms. u is set as $u = \beta \|\boldsymbol{\mu}^{c,m,t} - \boldsymbol{\mu}^{c,m,0:t-1}\|_2^2$ with $\boldsymbol{\mu}^{c,m,t}$ being the mean vector of $\mathbf{X}^{c,m,t}$. We measure the deviation between the current mini-batch and the past data by the Euclidean distance between their mean values. If the deviation is larger, the stronger constraint is expected to preserve the knowledge learned from the past data. β is a constant to adjust the magnitude of the deviation.

B. Model Learning and Classification

This section shows the details of the satellite target recognition method based on OnFA-KL model, the whole recognition scheme is depicted in Fig. 2, where D_t is the model updated from the t th training mini-batch.

Our satellite target recognition method based on OnFA-KL model has two stages, including training and classification, and some pre-processing techniques [22] should be applied in dealing with the abovementioned sensitivity problems.

In the training phase, we aim to calculate the posterior distributions of the model parameters based on the current mini-batch, instead of all data. As shown in formula (4), the objective function is composed of the lower bound on mini-batch and the KL divergence between global variables' posterior distributions given adjacent mini-batches.

The parameters of OnFA-KL model can be effectively inferred via SVB after t mini-batches, and the iterative updating formulas can be derived as follows.

1) Due to $\mathbf{s}_i^{c,m,t}$ being a local variable, the updating formulas are as follows:

$$\begin{aligned} q(\mathbf{s}_i^{c,m,t}) &= N(\mathbf{s}_i^{c,m,t} | \mathbf{m}_{\mathbf{s}_i^{c,m,t}}, \Sigma_{\mathbf{s}_i^{c,m,t}}) \\ \Sigma_{\mathbf{s}_i^{c,m,t}} &= (\mathbf{I}_K + \langle (\mathbf{A}^{c,m,t})^T \boldsymbol{\gamma}^{c,m,t} \mathbf{A}^{c,m,t} \rangle)^{-1} \\ \mathbf{m}_{\mathbf{s}_i^{c,m,t}} &= \Sigma_{\mathbf{s}_i^{c,m,t}} \langle (\mathbf{A}^{c,m,t})^T \boldsymbol{\gamma}^{c,m,t} \mathbf{x}_i^{c,m,t} \rangle. \end{aligned} \quad (5)$$

2) Loading matrix $\mathbf{A}^{c,m,t}$ is a global variable, and the updating formulas of $\mathbf{a}_j^{c,m,t}$ are as follows:

$$\begin{aligned} q(\mathbf{a}_j^{c,m,t}) &= N(\mathbf{a}_j^{c,m,t} | \mathbf{m}_{\mathbf{a}_j^{c,m,t}}, \Sigma_{\mathbf{a}_j^{c,m,t}}) \\ \Sigma_{\mathbf{a}_j^{c,m,t}} &= \left(\frac{1}{1+u} \langle \boldsymbol{\gamma}_j^{c,m,t} \rangle \sum_{i=1}^{N_t} \langle \mathbf{s}_i^{c,m,t} (\mathbf{s}_i^{c,m,t})^T \rangle + (\Sigma_{\mathbf{a}_j^{c,m,t-1}})^{-1} \right)^{-1}, \\ \mathbf{m}_{\mathbf{a}_j^{c,m,t}} &= \Sigma_{\mathbf{a}_j^{c,m,t}} \left(\frac{1}{1+u} \langle \boldsymbol{\gamma}_j^{c,m,t} \rangle \sum_{i=1}^{N_t} x_{ji}^{c,m,t} \langle \mathbf{s}_i^{c,m,t} \rangle \right. \\ &\quad \left. + (\Sigma_{\mathbf{a}_j^{c,m,t-1}})^{-1} \mathbf{m}_{\mathbf{a}_j^{c,m,t-1}} \right) \end{aligned} \quad (6)$$

where $x_{ji}^{c,m,t}$ is the j th element of $\mathbf{x}_i^{c,m,t}$.

3) $\boldsymbol{\gamma}_j^{c,m,t} \sim Ga(a_{\boldsymbol{\gamma}_j^{c,m,t}}, b_{\boldsymbol{\gamma}_j^{c,m,t}})$ is assumed, and the updating formulas of global variable $\boldsymbol{\gamma}_j^{c,m,t}$ are as follows:

$$\begin{aligned} a_{\boldsymbol{\gamma}_j^{c,m,t}} &= a_{\boldsymbol{\gamma}_j^{c,m,t-1}} + \frac{N_t}{2(1+u)} \\ b_{\boldsymbol{\gamma}_j^{c,m,t}} &= b_{\boldsymbol{\gamma}_j^{c,m,t-1}} + \frac{1}{2(1+u)} \\ &\quad \times \sum_{i=1}^{N_t} \left\langle \left(x_{ji}^{c,m,t} - (\mathbf{a}_j^{c,m,t})^T \mathbf{s}_i^{c,m,t} \right)^2 \right\rangle. \end{aligned} \quad (7)$$

In the experiments of this paper, the algorithm is terminated when the variation rate of the objective function $L(\boldsymbol{\theta}, \boldsymbol{\xi}^t)$ between two iterations is smaller than 10^{-6} .

In real scenarios, the satellite target data arrive in order of circles as shown in Fig. 3, where the red curves represent the three orbits of the satellite and the orbit segments that can be observed by radar are in yellow color. This shows that different circles may have different ranges of aspect angles (aspect-frames). Assuming that an initial model D_0 has been trained via OffFA model from the 0th training mini-batch. The derivation of updating formulas of the variational posteriors in the OffFA model can be summarized as: with $p(\mathbf{A}^{c,m,0})$ and $p(\boldsymbol{\gamma}^{c,m,0})$ being the prior distribution of $\mathbf{A}^{c,m,0}$ and $\boldsymbol{\gamma}^{c,m,0}$ respectively. If new data (the 1st training mini-batch) belong to existing aspect-frames, we update the existing model D_0 via equations (5)–(7); otherwise we construct new submodels for new aspect-frames via equations (8)–(10), shown at the bottom of the next page. Hence, we could get the updated model D_1 . As more training data are obtained, the model is updated according to this procedure. The streaming training process of

Algorithm 1 Training Steps of the OnFA-KL Model

Input: $\mathbf{X} = \{\mathbf{X}^{c,m,t}\}_{c=1,m=1,t=0}^{C,M_c,\infty}$, a_0, b_0 , and the threshold of the objective function variation rate ε

Output: $\{\mathbf{A}^{c,m,t}, \mathbf{S}^{c,m,t}, \boldsymbol{\gamma}^{c,m,t}\}_{c=1,m=1}^{C,M_c}$

```

1: for  $t = 0$  do
2:   Estimate  $\{\mathbf{A}^{c,m,0}, \mathbf{S}^{c,m,0}, \boldsymbol{\gamma}^{c,m,0}\}_{c=1,m=1}^{C,M_c}$  from
     equations(8)–(10);
3: end for
4: for  $t = 1, 2, \dots$  do
5:   for  $c = 1$  to  $C$  do
6:     for  $m = 1$  to  $M_c$  do
7:       if the submodel of  $m$ th frame of target  $c$  exists
8:         Initialize  $v = 0$ ,  $L(v) = -\infty$ ;
9:         Repeat
10:           $v = v + 1$ ;
11:          Estimate the parameters
              $\{\mathbf{s}_i^{c,m,t}\}_{i=1}^{N_t}, \{\mathbf{a}_j^{c,m,t}\}_{j=1}^P, \{\boldsymbol{\gamma}_j^{c,m,t}\}_{j=1}^P$ 
             from equations(5)–(7);
12:          Compute the lower bound  $L(v)$ ;
13:          Until  $L(v) - L(v-1) < \varepsilon L(v-1)$ 
             ( $\varepsilon = 10^{-6}$ )
14:        else
15:          Construct the new submodel from equations
             (8)–(10);
16:        end if
17:      end for
18:    end for
19: end for

```

satellite target recognition method based on OnFA-KL model is summarized as follows (Algorithm 1).

In the classification stage, the class label corresponding to a test sample $\mathbf{x}^*$ can be predicted with maximum posterior criteria at time t :

$$\begin{aligned} \text{label}(\mathbf{x}^*) &= \arg \max_c p(c, m, t | \mathbf{x}^*) \propto \arg \max_c p(c, m) p(\mathbf{x}^* | c, m, t), \end{aligned} \quad (11)$$

where the prior $p(c, m)$ is usually assumed as uniform distribution. The frame-conditional likelihood of $\mathbf{x}^*$ at time t can be expressed as

$$\begin{aligned} p(\mathbf{x}^* | c, m, t) &= N(\boldsymbol{\mu}^{c,m,0:t}, \langle \mathbf{A}^{c,m,t} \rangle \boldsymbol{\Omega}^{c,m,t} \langle \mathbf{A}^{c,m,t} \rangle^T + \langle (\boldsymbol{\gamma}^{c,m,t})^{-1} \rangle), \end{aligned} \quad (12)$$

with

$$\boldsymbol{\Omega}^{c,m,t} = 1/N_t \sum_{i=1}^{N_t} \langle \mathbf{s}_i^{c,m,t} \rangle \langle \mathbf{s}_i^{c,m,t} \rangle^T. \quad (13)$$

In consideration of the assumption that the frame models of each target are independent, the class-conditional likelihood given target c can be expressed as $p(\mathbf{x}^* | c, t) = \max_{m \in M_c} p(\mathbf{x}^* | c, m, t)$. According to Bayes theorem

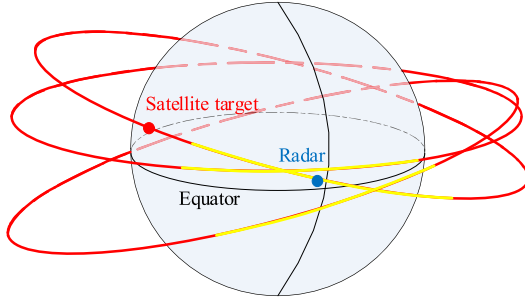


Fig. 3. Orbits of the satellite in radar coordinate system. The red curves represent the three circle orbits and the orbits segments that can be observed by radar are in yellow color.

$$p(c|\mathbf{x}^*, t) = p(\mathbf{x}^*|c, t) p(c) / \sum_{c=1}^C p(\mathbf{x}^*|c, t) p(c),$$
 the class membership of $\mathbf{x}^*$ at time t can be assigned to the class with the maximum posterior probability, i.e., $c^* = \arg \max_c p(c, t|\mathbf{x}^*)$.

Compared with the offline learning method, our satellite target recognition method based on OnFA-KL model is in online learning mode and can process the incoming data in real time without revisiting the past data, which will improve the efficiency of learning and save the cost of storage.

IV. EXPERIMENTAL RESULTS

In this section, we evaluate the performance of our proposed method on the simulation data of satellite targets. All the following experiments are performed with non-optimized MATLAB codes, on a Dell PC with 16 GB RAM and intel i7 CPU@3.40 GHz.

A. Introduction of Experimental Data

Before we start our experiments, we will briefly introduce the simulation data used in the experiments. For the lack of access to real measured HRRP data of satellite targets, the proposed method is investigated with HRRP simulation data of five satellite targets generated via electromagnetic simulation software FEKO [39]. Considering the satellite targets' orbital transfer, the azimuth angle and pitch angle of the radar line-of-sight (LOS) are simulated the actual satellite targets. The parameters of radar and five satellite targets are listed in TABLE I.

On the one hand, the data before orbital transfer almost cover all target-aspect angles of the data after orbital transfer; on the other hand, since the change of orbital parameters caused by orbital transfer, there are certain differences between the data before and after orbital transfer. Therefore, in our experiments, to test the generalization performance of the

recognition method, multi-circle data before orbital transfer are taken as the training data stream in the order of circles where the first circle of each target is taken as the 0th mini-batch. In detail, a total of 17432 HRRP samples from Apollo, 13170 from BeiDou, 14988 from DongFangHong, 16041 from KH-11, and 14459 from Landsat7 are used as training data stream. And the data in one circle after orbital transfer are taken as the testing samples.

As discussed in Section III, the essential preconsideration is to overcome the problem of target-aspect, time-shift, and amplitude-scale sensitivity for radar target recognition. According to the scattering center model [36], the central issue of the variation of target-aspect is scatterer's motion through range cells (MTRC), i.e., target-aspect sensitivity, thereby changing substantially the statistical characteristics of HRRP. The aspect-frame partition method that avoids the MTRC of scatterers is commonly used to cope with this problem, and the range of target-aspect change without the occurrence of MTRC is defined as

$$\delta\varphi \leq (\delta\varphi)_{\text{MTRC}} = \Delta R/L, \quad (14)$$

where ΔR and L are the length of range resolution cell (namely the resolution of the range cell) and the cross length of the target, respectively. In the following experiments, the HRRP dataset of a target can be divided into several frames based on $(\delta\varphi)_{\text{MTRC}}$, and each frame in each circle has about 90 samples with the dimension being 512. After aspect-frame partition, the amplitude scale sensitivity [12], [16] should be considered. In practice, HRRPs measured under different conditions may have different amplitude-scales. Hence, each HRRP sample should be normalized by dividing its L_2 norm to overcome the amplitude-scale sensitivity problem. Furthermore, to overcome the time-shift sensitivity problem [4] caused by the variation of target signal position in different HRRP samples, the slide correlation alignment [33] is used within each frame as the time-shift compensation techniques. Fig. 4 represents the HRRP samples of these five satellite targets. From Fig. 4 we can see that the echo amplitudes of range profiles are discriminative for five satellite targets.

B. Recognition Performance

The experiments of our satellite target recognition method are based on above-introduced simulation data.

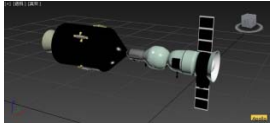
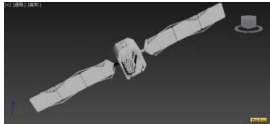
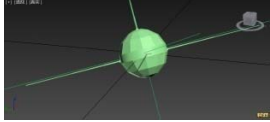
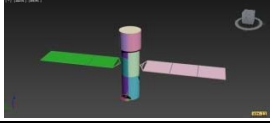
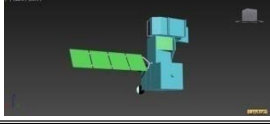
In our experiment settings, firstly, an initial model D_0 is trained on the 0th mini-batch, i.e., the first circle of observed data, using OffFA model; secondly, the rest of training data stream is used to update the current model via OnFA-KL model to get $D_1, D_2, \dots$; if the data of some targets run

$$q(\mathbf{s}_i^{c,m,0}) \propto \exp\left(\left\langle \ln p(\mathbf{x}_i^{c,m,0}, \mathbf{s}_i^{c,m,0} | \mathbf{A}^{c,m,0}, \boldsymbol{\gamma}^{c,m,0}) \right\rangle_{q(\mathbf{A}^{c,m,0})q(\boldsymbol{\gamma}^{c,m,0})}\right), \quad (8)$$

$$q(\mathbf{A}^{c,m,0}) \propto p(\mathbf{A}^{c,m,0}) \exp\left(\left\langle \ln p(\mathbf{X}^{c,m,0}, \mathbf{S}^{c,m,0} | \mathbf{A}^{c,m,0}, \boldsymbol{\gamma}^{c,m,0}) \right\rangle_{q(\mathbf{S}^{c,m,0})q(\boldsymbol{\gamma}^{c,m,0})}\right), \quad (9)$$

$$q(\boldsymbol{\gamma}^{c,m,0}) \propto p(\boldsymbol{\gamma}^{c,m,0}) \exp\left(\left\langle \ln p(\mathbf{X}^{c,m,0}, \mathbf{S}^{c,m,0} | \mathbf{A}^{c,m,0}, \boldsymbol{\gamma}^{c,m,0}) \right\rangle_{q(\mathbf{S}^{c,m,0})q(\mathbf{A}^{c,m,0})}\right) \quad (10)$$

TABLE I
PARAMETERS OF RADAR AND FIVE SATELLITE TARGETS IN THE EXPERIMENTS

Radar Parameters	Center Frequency		16.67GHz		
	Pulse Repetition Frequency		10Hz		
	Sampling Frequency after Dechirp		10MHz		
	Bandwidth		2GHz		
Radar Location	39.9°N 116.4°E				
CAD Models	Real Images	Satellite Targets	Length(m)	Width(m)	Height(m)
		Apollo	21.6	9.6	5.1
		BeiDou	9.4	1.2	2.0
		DongFangHong	5.5	5.5	2.0
		KH-11	30.7	4.0	12.0
		Landsat7	7.6	2.7	3.9

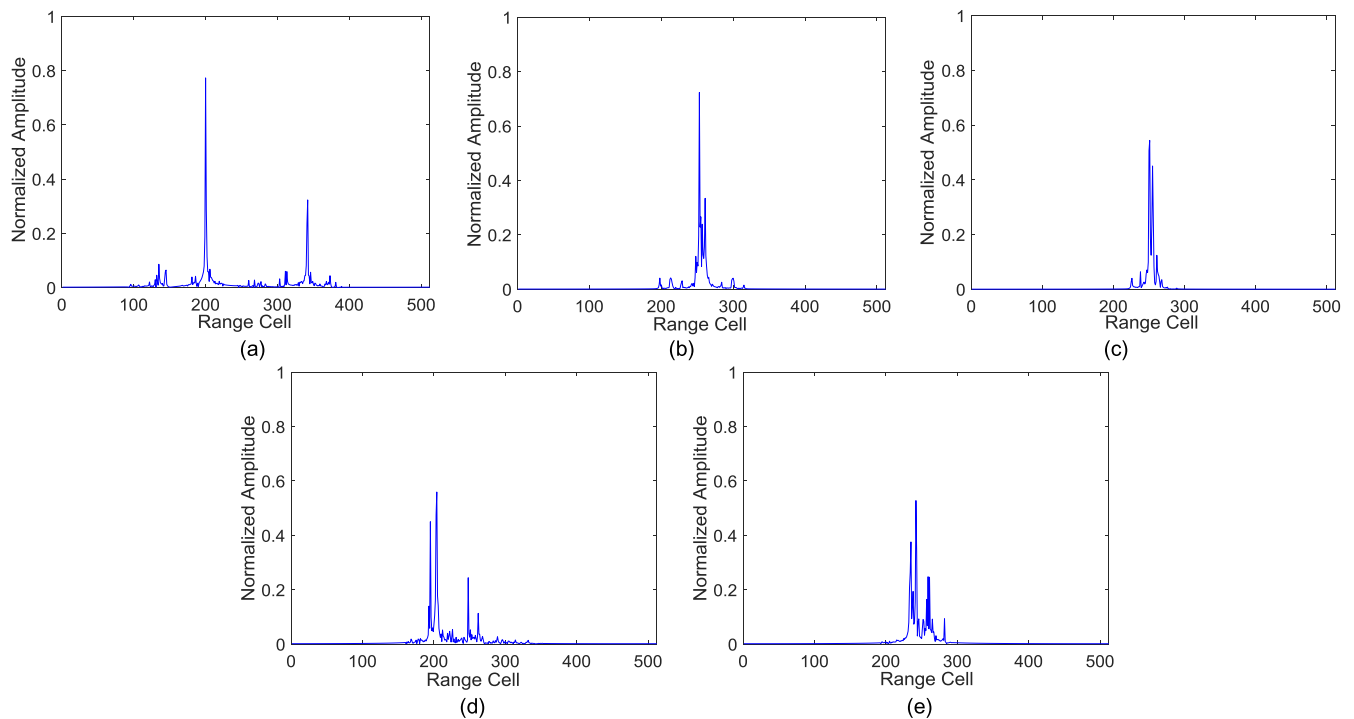


Fig. 4. HRRP sample examples of satellite targets. (a) HRRP sample example of Apollo; (b) HRRP sample example of BeiDou; (c) HRRP sample example of DongFangHong; (d) HRRP sample example of KH-11; (e) HRRP sample example of Landsat7.

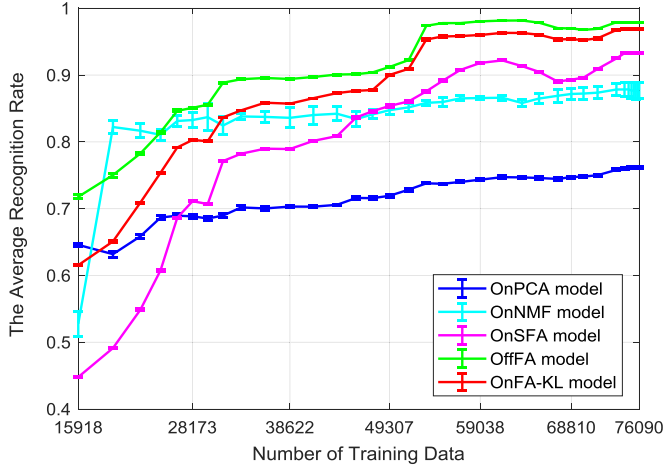


Fig. 5. The variation of average recognition rates with the number of training data.

out in the later stages, the submodels of the rest targets still be updated. Moreover, we evaluate our method on testing samples every 10 frames' updates of each target. In order to further illustrate the performance of our method, the following experiments are repeated 10 times, and the results are averaged over the 10 times' results. Meanwhile, the standard deviations are also calculated and presented by error bars to describe the uncertainty caused by random initialization of parameters.

In the following, to verify the validity of our OnFA-KL model, we first evaluate the recognition performance by comparing with OnPCA model [17], OnNMF model [38], OnSFA model [18], and OffFA model. For OnSFA model, the parameters are updated via SVI which is commonly used for large-scale datasets. Here, we use OnSFA model in streaming mode. The recognition performance is measured by the average recognition rate. The average recognition rate ARR can be calculated as:

$$ARR = \frac{1}{C} \left(\sum_{c=1}^C \frac{R_c^{test}}{N_c^{test}} \right) \quad (15)$$

where R_c^{test} and N_c^{test} are the number of correctly classified samples and the total samples of target c in testing data, respectively. The recognition rates of our OnFA-KL model and other related models are shown in Fig. 5, whose horizontal axis is the number of training data and vertical axis represents the variation of the average recognition rate. Note that the 1st value corresponds to testing the initial model D_0 .

Fig. 5 depicts that the average recognition rates of the above-mentioned models increase with the number of training data (with the arrival of training data). In the beginning, the average recognition rate is low because of lacking the accurate description of HRRP data with less training data. Later, the average recognition rate increases gradually and then tends to be stable finally. The reason is that the model parameter estimation becomes more and more accurate with the number of training data increases. The recognition performance of OnPCA model and OnNMF model, which transfer the data into the low-dimension subspace, is lower than that

of OnFA-KL model because of lacking the accurate description for HRRP data. OnSFA model is commonly used for large-scale datasets, where the data should be processed in random sequence to gain the generalization; therefore, our OnFA-KL model outperforms OnSFA model in streaming mode. Furthermore, to analyze the recognition performance for our OnFA-KL model, we compare the performance of the OnFA-KL model to its corresponding offline model (OffFA model). In the beginning, the recognition performance of OffFA model is better than that of OnFA-KL model. With the arrival of training data, the performance difference between them becomes smaller. Finally, the performance of OnFA-KL model approaches that of OffFA model.

Moreover, as an online learning model, the computation burden is significant to evaluate the practicability of the model in real applications. In Fig. 6, we analyze the storage cost and time consumption of our OnFA-KL model and OffFA model with the number of training data (with the arrival of training data). In our OnFA-KL model, we only need to store the current training data and the related model parameters, while all historical training data and the related model parameters need to be stored in OffFA model. Thus, the storage cost of our OnFA-KL model is less than that of OffFA model. Since only current training data are used for updating our OnFA-KL model instead of learning from scratch, the time consumption of our OnFA-KL model is much less than that of OffFA model.

C. Analysis of Our Proposed Method

In our proposed method, the KL constraint term is crucial. We expect the KL constraint term to assist in preserving the previous learned knowledge into our OnFA-KL model with the arrival of streaming training data. Therefore, we make an analysis on the KL constraint term. The key factor of the KL constraint term is the setting of its weight value $u(\beta)$. In our experiments, we use cross-validation [40] to dynamically search the optimal β . We make an analysis on the degree of the previous learned knowledge preservation of the 0th mini-batch with the arrival of new data with or without KL constraint term and compare their performance in the following.

The mean of logarithm likelihoods of training data represents the data description ability of the statistical model; thus, the evaluation is based on the mean of logarithm likelihoods of the 0th mini-batch $\{\mathbf{X}^{c,m,0}\}_{c=1, m=1}^{C, M_c}$. Fig. 7 shows the mean of logarithm likelihoods of the 0th mini-batch with or without KL constraint term. In the following, an online FA model without KL constraint term (i.e., $\beta = 0$) is referred to as OnFA model. The horizontal axis is the number of training data and the vertical axis represents the variation of the mean of logarithm likelihoods. We can see that: 1) with the arrival of the data, the mean of logarithm likelihoods of the 0th mini-batch gradually decreases, which represents the problem of catastrophic forgetting; 2) with KL constraint term, the mean of logarithm likelihoods decreases less, i.e., the previous learned knowledge can be preserved. Therefore, the KL constraint term can alleviate catastrophic forgetting.

Moreover, we also analyze the recognition performance of OnFA-KL model and OnFA model, and the average

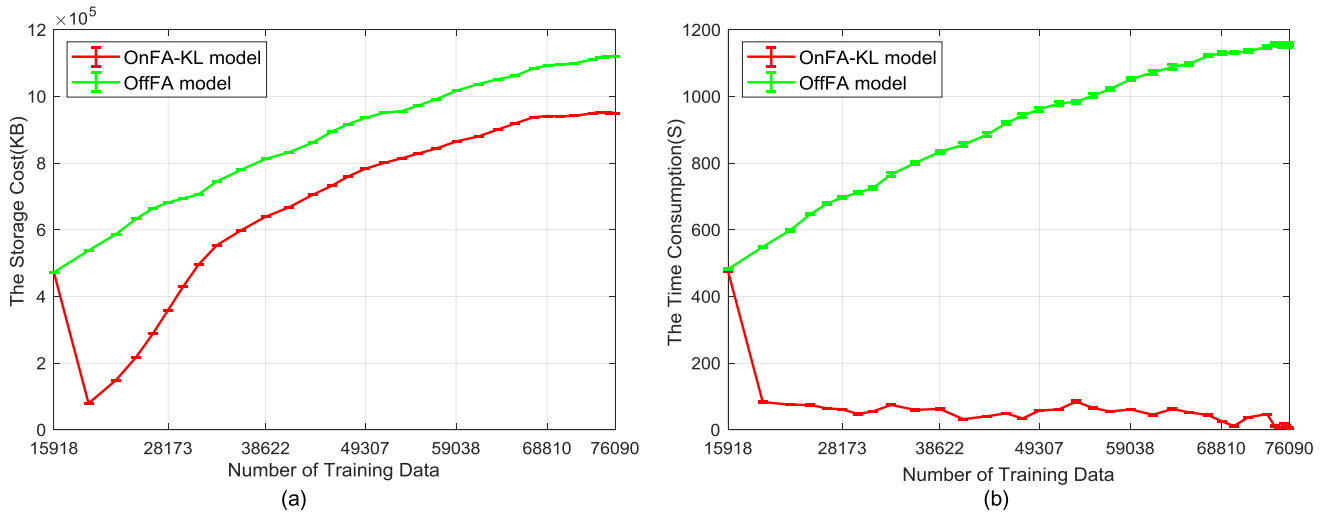


Fig. 6. The variation of computation burden with the number of training data (with the arrival of training data). (a) Storage cost; (b) Time consumption.

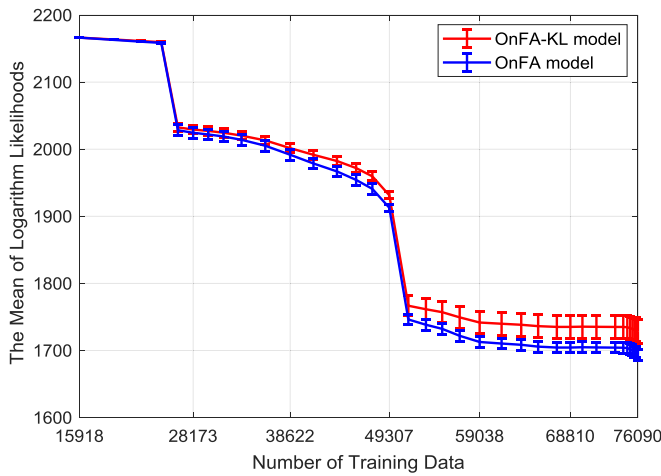


Fig. 7. The mean of log likelihoods of OnFA-KL model and OnFA model with the number of training data.

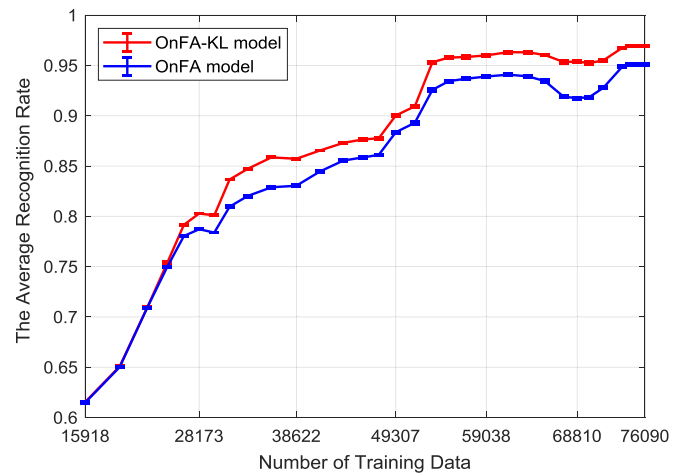


Fig. 8. The variation of average recognition rates of OnFA-KL model and OnFA model with the number of training data.

recognition rates are shown in Fig. 8. We can see that: 1) with the arrival of the data, the average recognition rates gradually increase; 2) our OnFA-KL model achieves better recognition performance than OnFA model. Hence, adding the KL constraint term for alleviating catastrophic forgetting can improve the recognition performance.

Besides the KL constraint term, the batch size which represents the number of new samples of each frame every update, is another crucial factor. Fig. 9 presents the variation of average recognition rates of our OnFA-KL model with the arrival of the training data under different batch sizes. We could observe that the final average recognition rates become larger with the increase of batch size. The reason is that if we choose the batch size too small, for example, batch size = 5, each mini-batch would not provide sufficient samples for updating the parameters. Moreover, we also analyze the final (maximum) storage cost and the total time consumption under different batch sizes in TABLE II, where the values in the parentheses represent the standard deviations. With the increase of the batch size, the final storage cost becomes

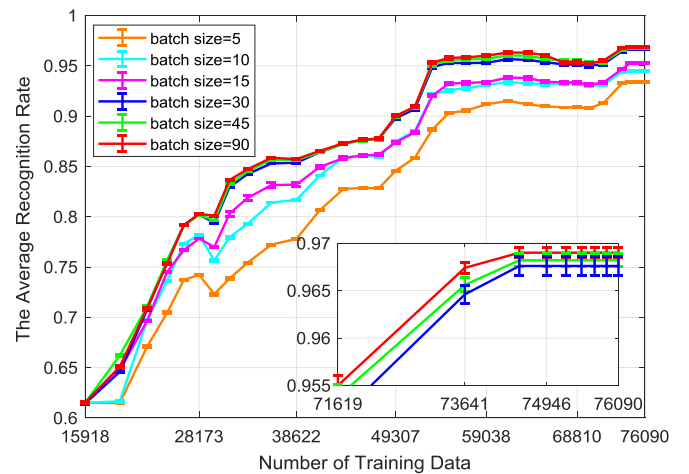


Fig. 9. The variation of average recognition rates of the OnFA-KL model under different batch sizes with the number of training data.

larger, whereas the total time consumption decreases. The main reason is that the more current training data are needed

TABLE II
THE FINAL STORAGE COST AND TIME CONSUMPTION
UNDER DIFFERENT BATCH SIZES

Batch size	The final storage cost (KB)	The total time consumption (S)
5	9.1820×10^5 (26.74)	3.3357×10^3 (13.71)
10	9.4436×10^5 (9.59)	2.9225×10^3 (2.55)
15	9.4895×10^5 (13.31)	2.5813×10^3 (3.49)
30	9.4961×10^5 (8.93)	2.3242×10^3 (4.32)
45	9.4995×10^5 (13.21)	2.1811×10^3 (47.54)
90	9.5028×10^5 (9.72)	2.0702×10^3 (5.65)

to be stored and the fewer update times are required to train data of the same size. Therefore, we should use the data as much as possible within the storage requirements to achieve better recognition performance.

V. CONCLUSION

We have developed a novel online satellite target recognition method based on OnFA-KL model. In our proposed OnFA-KL model, with the arrival of new data, the parameters of the OnFA-KL model can be updated in online mode without revisiting the past data. Moreover, a KL divergence term is introduced to preserve the previous learned knowledge of the past data for alleviating catastrophic forgetting. Firstly, the experiments on satellite target simulation data show that our OnFA-KL model can achieve comparable recognition performance with higher learning efficiency and lower storage compared with the corresponding offline method. Secondly, we validate the catastrophic forgetting problem can be effectively alleviated by adding KL divergence term.

However, since our OnFA-KL model is a frame-independent model, the storage of model parameters increases as the number of frames increases. In addition, the weight value of the KL divergence term cannot be learned adaptively through the data learning process. In our future work, we are interested in investigating an online model, of which the storage of model parameters is fixed. Besides this, a more effective and adaptive method should be developed to alleviate catastrophic forgetting problem.

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